Sets 3 — Probability from Venn diagrams · Conditional probability

Marks in brackets. ★ = harder question. $P(A) = \dfrac{n(A)}{n(\mathscr{E})}$; $P(A \mid B)$ = probability of A *given* B. Give probabilities as fractions in simplest form.

1 In a class of 30 students, 21 passed a Physics test (P) and 24 passed a Chemistry test (C). 2 students failed both. [5]

a Draw a Venn diagram.

b A student is chosen at random. Find **i** $P(\mathrm{P})$ **ii** $P(\mathrm{P} \cap \mathrm{C})$ **iii** $P(\mathrm{P} \cup \mathrm{C})$ **iv** $P(\mathrm{P}' \cap \mathrm{C})$.

a Both $= x$: $(21 - x) + x + (24 - x) + 2 = 30 \Rightarrow x = 17$. Regions: P only 4, both 17, C only 7, neither 2.
b i $\dfrac{21}{30} = \dfrac{7}{10}$ **ii** $\dfrac{17}{30}$ **iii** $\dfrac{4 + 17 + 7}{30} = \dfrac{28}{30} = \dfrac{14}{15}$ **iv** C but not P: $\dfrac{7}{30}$.

2 A café records what 150 customers buy one morning: 120 buy a coffee, 70 buy a cake, and every customer buys at least one of the two. A customer is picked at random. Find the probability that they bought [5]

a a coffee and a cake **b** a coffee or a cake but not both.

Both $= 120 + 70 - 150 = 40$ (nobody is outside). Coffee only 80, cake only 30. **a** $\dfrac{40}{150} = \dfrac{4}{15}$. **b** $\dfrac{80 + 30}{150} = \dfrac{110}{150} = \dfrac{11}{15}$.

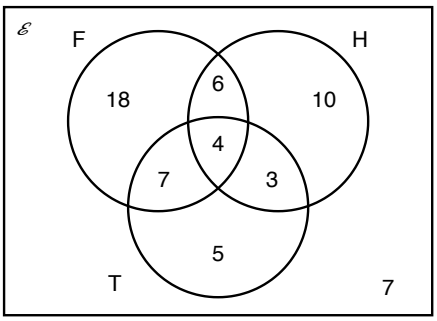
3 In a survey, 80% of students own a bike (B), 30% own a skateboard (S) and 15% own neither. A student is chosen at random. Find the probability that the student [5]

a owns a bike or a skateboard but not both

b owns a skateboard, given that they own a bike.

Work in percentages: both $= 80 + 30 - (100 - 15) = 25\%$. Bike only 55%, skateboard only 5%, neither 15%. **a** $\dfrac{55 + 5}{100} = \dfrac{3}{5}$. **b** Restrict to bike owners (80%): $P(\mathrm{S} \mid \mathrm{B}) = \dfrac{25}{80} = \dfrac{5}{16}$.

4 The Venn diagram shows which of football (F), hockey (H) and tennis (T) the students in a year group play. [5]



a How many students are in the year group?

b A student is chosen at random. Find **i** $P(\mathrm{H})$ **ii** $P(\mathrm{F} \cap \mathrm{H} \cap \mathrm{T})$ **iii** $P(\mathrm{T} \mid \mathrm{H})$.

a $18+6+10+7+4+3+5+7=60$. **b i** $n(\mathrm{H})=6+10+4+3=23$: $\dfrac{23}{60}$. **ii** $\dfrac{4}{60}=\dfrac{1}{15}$. **iii** Given H, only the 23 hockey players count; of them $4+3=7$ play tennis: $\dfrac{7}{23}$.

5 A music club has 40 members, each of whom plays at least one of piano (P), guitar (G) and drums (D). 24 play piano, 20 play guitar and 15 play drums; 9 play piano and guitar, 7 play guitar and drums, 8 play piano and drums. A member is chosen at random. Find **a** $P(\mathrm{P} \cap \mathrm{G} \cap \mathrm{D})$ **b** $P(\mathrm{D} \mid \mathrm{G})$. [5]

Let x play all three. Pairs only: $9-x$, $7-x$, $8-x$. Singles: P only $24-(9-x)-(8-x)-x=7+x$, G only $20-(9-x)-(7-x)-x=4+x$, D only $15-(7-x)-(8-x)-x=x$. Total: $(7+x)+(4+x)+x+(9-x)+(7-x)+(8-x)+x=35+x=40 \Rightarrow x=5$.

a $\dfrac{5}{40}=\dfrac{1}{8}$. **b** Among the 20 guitarists, those who also play drums are $(7-x)+x=7$: $P(\mathrm{D} \mid \mathrm{G})=\dfrac{7}{20}$.

Total: 25 marks

Check: $P(A \cup B)$ counts the overlap once · "but not both" = the two outer parts · $P(A \mid B)=\dfrac{n(A \cap B)}{n(B)}$ — the denominator is the *given* set, not the whole universal set · percentages work like counts out of 100 · simplify every fraction.